High Level Architecture

Trusted Executive Graph Platform

Architecture Document

Version: 1.0
Architecture Owner: Platform Architecture
Purpose: Build an AI-driven platform that creates **trusted executive profiles, endorsements, and reputation scoring using a trust graph and agentic AI.**

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1. Executive Summary

Traditional professional platforms like LinkedIn rely heavily on **self-reported profiles and low-trust endorsements**. Executive hiring and reputation assessment often depend on **informal networks and references** rather than platform signals.

This platform introduces a **Trusted Executive Graph**, combining:

- Verified executive profiles
- High-trust endorsements
- Graph-based relationship analysis
- AI agents for reputation intelligence
- Executive trust scoring

The system leverages **agentic AI** through Moltbot to automate tasks such as:

- Profile verification
- Endorsement discovery
- Reputation scoring
- Network analysis

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2. Goals and Objectives

Primary Goals

1. Build a **trusted executive reputation platform**
2. Establish **verified endorsements from credible executives**
3. Implement **trust graph relationships**
4. Provide **AI-driven insights through agents**
5. Generate **trust scores for executives**

Non-Goals (MVP)

- Full recruiting platform
- Resume parsing marketplace
- Social networking platform

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3. Key System Capabilities

Capability	Description
Executive Profile	Verified leadership profile
Endorsements	Trust-weighted executive endorsements
Trust Graph	Graph relationships between executives
Trust Score	Credibility score based on signals

Agentic Insights AI recommendations and network insights

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4. High Level Architecture

```
+-----------------------------------------------------+
|                Frontend Dashboard                   |
|            (React / Angular UI)                     |
+-----------------------------------------------------+
                  |
                  |
+-----------------------------------------------------+
|                API Gateway Layer                    |
|                 Node.js / Express                   |
+-----------------------------------------------------+
                  |
      +-------------------------------+
      |                               |
+---------------+             +----------------------+
| AI Agent Layer |            | Business Services  |
| (Moltbot)      |            | Profile / Endorse  |
+---------------+             +----------------------+
      |                               |
      |                               |
+---------------------+      +----------------------+
| Trust Graph Engine  |      | Profile Database     |
| Neo4j Graph DB      |      | PostgreSQL           |
+---------------------+      +----------------------+
```

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5. Logical Architecture Components

5.1 Frontend Layer

User interface for:

- Executive profiles
- Trust score dashboards

- Endorsement management
- Network visualization

Technologies

- React or Angular
- Graph visualization libraries

Capabilities

- Trust graph visualization
- Profile analytics
- AI insight display

5.2 API Layer

Backend APIs handle:

- Profile creation
- Endorsement management
- Trust score retrieval
- Agent orchestration

Technology

- Node.js
- Express REST APIs

Example endpoints

POST /executives
POST /endorsements
GET /trust-score/{id}
GET /network/{id}

5.3 AI Agent Layer

The AI layer runs agent workflows using
Moltbot.

Agents automate reputation and network analysis.

Agents

Profile Agent

Responsibilities:

- Normalize executive profiles
- Extract leadership signals
- Validate credentials

Endorsement Agent

Responsibilities:

- Identify potential endorsers
- Generate endorsement requests
- Update trust graph

Trust Score Agent

Responsibilities:

- Compute executive credibility score
- Explain score reasoning
- Provide recommendations

Network Intelligence Agent (future)

Responsibilities:

- Discover hidden connections
- Suggest introductions
- Identify board opportunities

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6. Data Architecture

The platform uses **two types of data stores**.

6.1 Relational Database

Technology

- PostgreSQL

Purpose

Store structured profile data.

Tables

Executives
Endorsements
Organizations

Publications
Conferences

Example Schema

```
Executives
---------
id
name
title
company
industry
linkedin_url
verified
created_at
```

6.2 Graph Database

Technology

- Neo4j

Purpose

Represent relationships between executives.

Example relationships

Executive → ENDORSES → Executive
Executive → WORKED_WITH → Executive
Executive → MENTORED → Executive

Graph benefits

- Trust propagation
- Network discovery
- Centrality analysis

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7. Trust Graph Model

Example Graph

```
CEO A
  |
ENDORSES
  |
VP B
  |
WORKED_WITH
  |
Executive C
```

Each edge contains metadata:

```
relationship_type
trust_weight
verification_status
time_period
```

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8. Trust Score Algorithm

Trust score is derived from multiple signals.

```
Trust Score =
 endorsement_weight
+ leadership_roles
+ external_recognition
+ network_centrality
```

Example scoring weights

Signal	Weight
Executive endorsement	10
Leadership role	5
Publication	3

Conference speaking	4

Score range

0 – 100

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9. Agent Workflow

Typical user interaction.

User creates profile
↓
Profile Agent analyzes credentials
↓
Endorsement Agent suggests endorsers
↓
Trust Graph updated
↓
Trust Score Agent computes credibility
↓
Dashboard displays insights

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10. Security Architecture

Security controls include:

Authentication

- OAuth integration
- corporate email verification

Authorization

Role-based access control

Roles

Executive
Endorser
Admin
Recruiter

Data Protection

- encrypted credentials
- secure API endpoints
- audit logs

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11. Scalability Considerations

Future scaling strategies.

Horizontal API scaling

Deploy backend services using containers.

Example platforms

- Amazon Web Services
- Google Cloud

Graph scaling

Use cluster deployments of:

- Neo4j Enterprise

Agent scaling

Agent workers can run independently using:

- containerized Moltbot workers

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12. Deployment Architecture

Example deployment

Cloud Infrastructure

```
Load Balancer
   |
API Service (Node)
   |
Agent Service (Python Moltbot)
   |
Postgres Cluster
Neo4j Cluster
```

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13. Observability

System monitoring includes:

Logging

- application logs
- agent activity logs

Metrics

- API latency
- agent execution time

- trust score calculations

Tracing tools

- OpenTelemetry

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14. Risks and Mitigation

Risk	Mitigation
Fake endorsements	verification workflows
Identity spoofing	domain email verification
Trust score manipulation	graph validation rules
Agent hallucinations	deterministic scoring

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15. Future Enhancements

Future capabilities include:

Executive network intelligence

Predict influential relationships.

Board opportunity matching

Match executives with boards.

Reputation monitoring

Monitor:

- conferences
- publications
- industry recognition

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16. Business Impact

The platform enables:

- trusted executive reputation
- faster executive hiring
- board recruitment
- network intelligence

Potential users

- executives
- venture capital firms
- board recruiters
- executive search firms

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17. Conclusion

This architecture introduces a **next-generation executive reputation system** combining:

- verified signals

- graph analytics
- AI agents
- trust scoring

The integration of **agentic AI using Moltbot** enables automated reputation analysis and network discovery, significantly improving the reliability of professional credibility signals.

C4 Architecture

C4 Architecture – Trusted Executive Graph Platform

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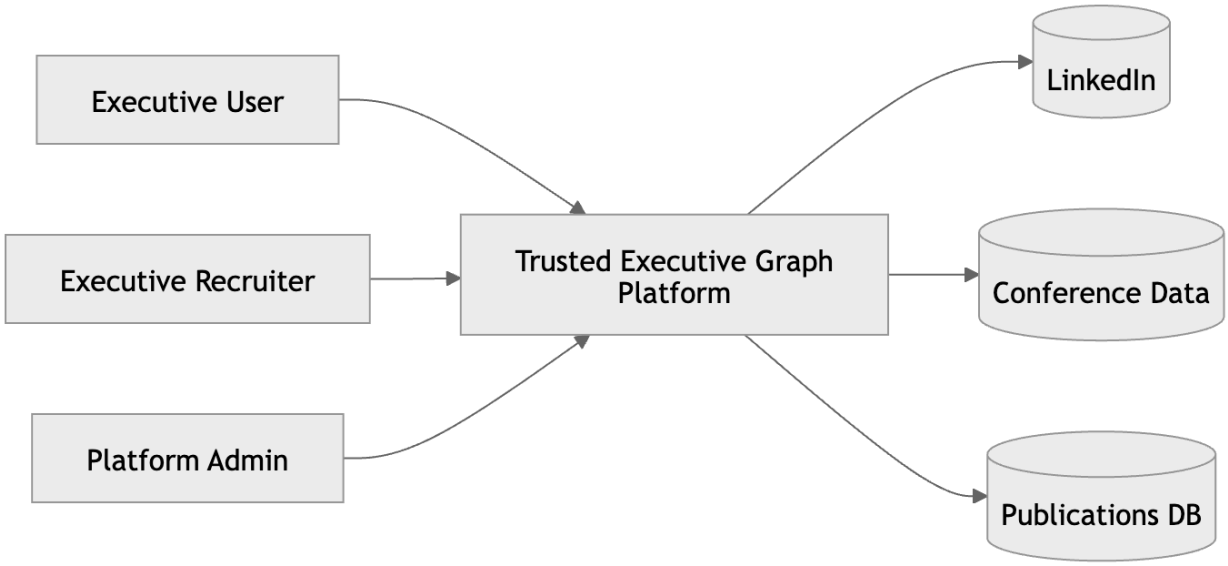
1. Level 1 – System Context Diagram

This shows **who interacts w**



ith the system.

Actors

Executive

- Creates trusted profile
- requests endorsements
- views trust score

Recruiter

- evaluates executive credibility
- searches trusted candidates

Admin

- verifies profiles
- monitors endorsements

External systems include:

- LinkedIn profile signals
- conference databases such as IEEE events

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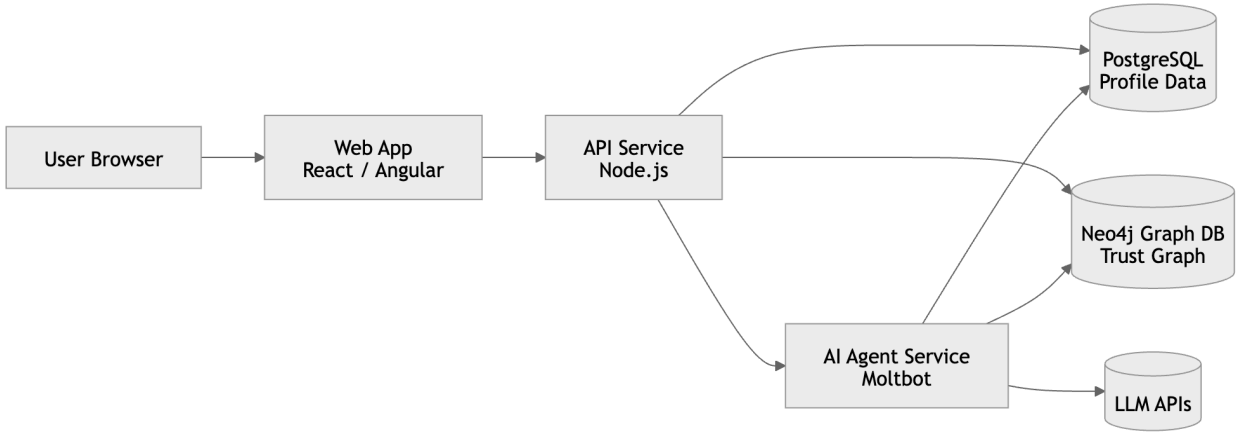
2. Level 2 – Container Diagram

This shows **major s**



ervices and technologies.

Containers

Web Application

- UI for executive dashboards
- trust graph visualization

API Service

- profile management
- endorsement APIs
- trust score retrieval

Agent Service

Uses **Moltbot**

Responsibilities:

- profile verification
- endorsement discovery
- trust score explanation

Relational Database

Stores:

- executives
- endorsements
- publications

Graph Database

Uses Neo4j

Stores:

- executive relationships
- trust graph edges

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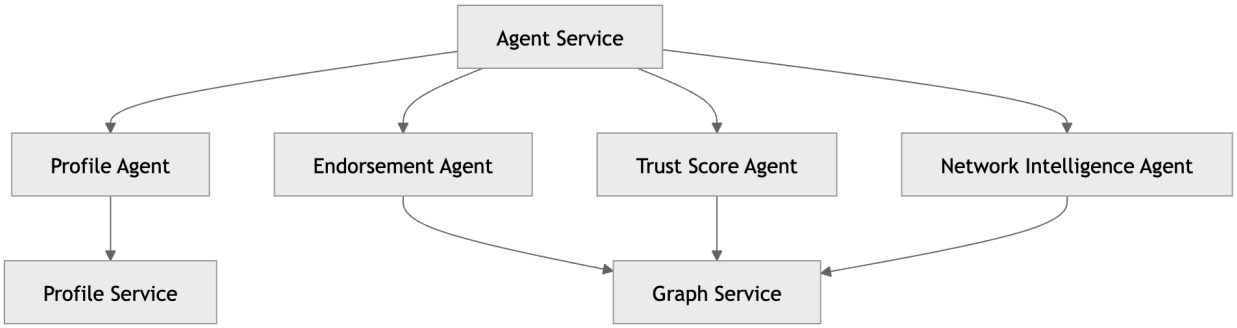
3. Level 3 – Component Diagram (Agent Service)



This diagram zooms into the **Agent Layer**.

Profile Agent

Responsibilities:

- normalize profile data
- extract leadership signals
- validate achievements

Endorsement Agent

Responsibilities:

- recommend endorsers
- generate endorsement requests
- update trust graph

Trust Score Agent

Responsibilities:

- calculate credibility score
- explain scoring rationale

Network Intelligence Agent

Responsibilities:

- analyze executive network
- suggest strategic connections
- identify board opportunities

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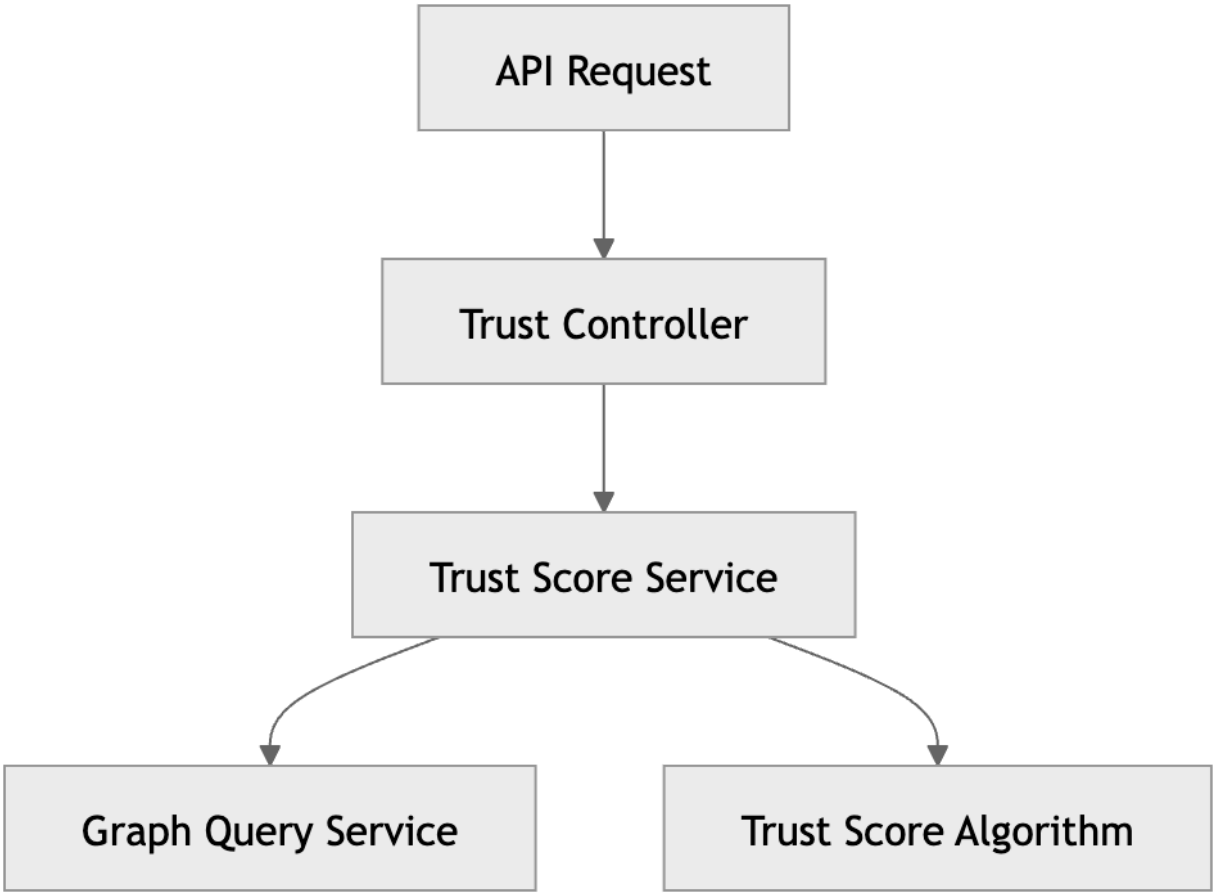
4. Level 4 – Code Diagram (Trust Score Agent Example)



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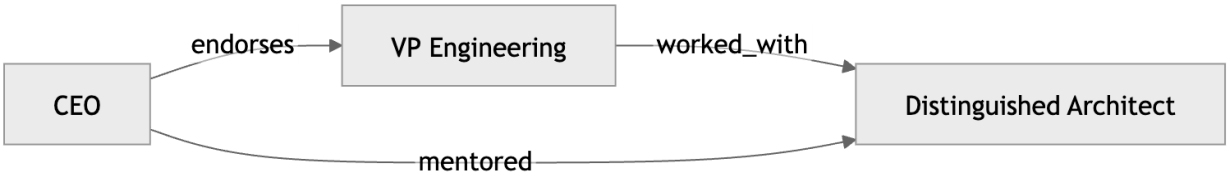
5. Trust Graph Example



Each relationship contains attributes:

trust_weight
relationship_type
verification_status

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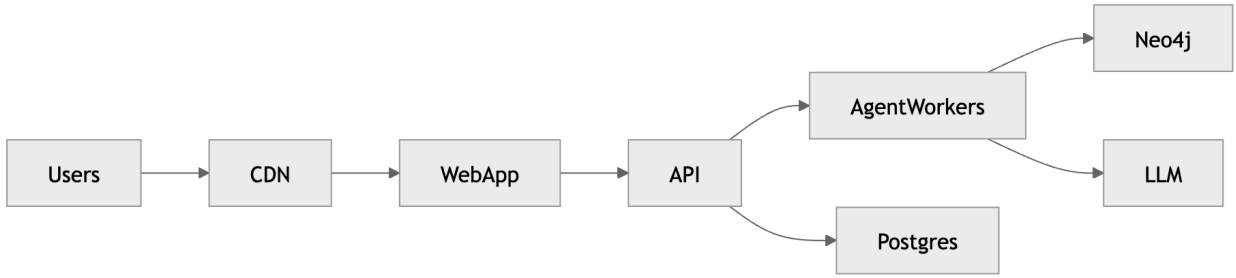
6. Deployment Diagram



Example cloud deployment:

Possible cloud providers:

- Amazon Web Services
- Google Cloud

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7. Architecture Strengths

This architecture provides:

Strong trust signals

Graph-based endorsements.

Agentic automation

Using **Moltbot**

Network intelligence

Graph analytics.

Scalable architecture

Microservice-ready design.